

Pre-Thesis II Report

Multi-Task Deep Learning Framework for Unified Cattle Health and Behavior Monitoring

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B.Sc. in Computer Science and Engineering

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Declaration

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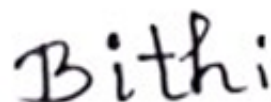
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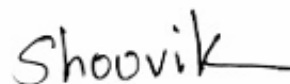
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Abstract

Precision livestock farming requires automated systems that can monitor animal health and welfare continuously without depending only on manual inspection. Existing cattle monitoring studies usually solve one task at a time, such as body condition scoring, behavior recognition, lameness detection, or cattle identification. This single-task design creates high computational cost, weak practical integration, and limited ability to reuse shared visual information across related cattle health tasks. This Pre-Thesis II work implements and evaluates a unified deep learning framework for four cattle monitoring objectives: Body Condition Scoring (BCS), behavior recognition, lameness detection, and individual cattle identification.

The proposed system uses public datasets and a shared computer vision backbone followed by a Convolutional Block Attention Module (CBAM) and task-specific prediction heads. The BCS head is trained as an ordinal regression problem using CORAL loss, the behavior head uses focal loss to address severe class imbalance, the lameness head uses binary classification with both spatial and spatiotemporal variants, and the ID head uses softmax classification. Five backbone architectures were trained as single-task baselines by different team members: ResNet-18, MobileNetV3-Small, ResNet-50, DenseNet121, and EfficientNetB0. The best average backbone was then used for the final multi-task implementation.

The single-task experiments show that MobileNetV3-Small achieved the best Dryad BCS test MAE of 0.5250, EfficientNetB0 achieved the best ScienceDB BCS test MAE of 0.5566, and EfficientNetB0 achieved the best behavior test Macro F1-score of 0.7445 on MmCows. The preliminary lameness model reached a test AUC of 0.9829 and a test accuracy of 93.05%, while the OpenCows2020 cattle ID model reached 86.49% test Top-1 accuracy. The final spatiotemporal multi-task model achieved 100.00% lameness accuracy with AUC 1.0000, 97.18% cow ID accuracy, BCS MAE of 0.7849, and behavior Macro F1-score of 0.4775. The results show that multi-task learning offers strong deployment advantages and improves temporal tasks, although static spatial tasks can suffer from shared-backbone interference. This report presents the complete methodology, implementation progress, baseline comparison, multi-task re-

sults, limitations, and future work for the final thesis stage.

Keywords: Multi-Task Learning, Cattle Health Monitoring, Body Condition Scoring, Behavior Recognition, Lameness Detection, Cattle Identification, CORAL Loss, Focal Loss, CBAM, EfficientNetB0, Spatiotemporal Deep Learning, Precision Livestock Farming

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Nomenclature

Abbreviation	Meaning
AI	Artificial Intelligence
AUC	Area Under the Receiver Operating Characteristic Curve
BCS	Body Condition Score / Body Condition Scoring
BCE	Binary Cross-Entropy
CBAM	Convolutional Block Attention Module
CBVD-5	Cow Behavior Video Dataset with five behavior classes
CE	Cross-Entropy
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CORAL	Consistent Rank Logits / Rank Consistent Ordinal Regression
CUDA	Compute Unified Device Architecture
DGE	Depth, Grayscale, and Edge channels
DL	Deep Learning
F1	Harmonic mean of precision and recall
GPU	Graphics Processing Unit
ID	Individual Cattle Identification
IoT	Internet of Things
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MTL	Multi-Task Learning
PLF	Precision Livestock Farming
RGB	Red, Green, Blue color image
RGB-D	RGB image with depth information
ROC	Receiver Operating Characteristic
ST	Spatiotemporal
YOLO	You Only Look Once object detection framework

Chapter 1

Introduction

1.1 Background

The livestock industry has become an important application area for artificial intelligence because animal health, welfare, and productivity depend on regular observation. Cattle monitoring is especially important for dairy and beef production, where a delay in detecting body condition decline, lameness, abnormal behavior, or identity mismatch can directly affect farm profitability and animal welfare. Traditional assessment depends on human experts who visually inspect animals, record observations, and decide whether intervention is needed. Although expert observation remains valuable, it is time-consuming, subjective, and difficult to apply continuously in large farms.

Body Condition Scoring (BCS) is a standard method for estimating the energy reserve and nutritional condition of a cow. It is normally assessed from visible or tactile features around the spine, ribs, hooks, pins, and tail-head region. Manual BCS is useful but varies between observers and cannot be performed continuously for every cow. Lameness is another major welfare concern because abnormal gait, pain, and foot disease reduce feed intake, reproductive performance, and milk yield. Behavior recognition is also important because changes in standing, walking, feeding, drinking, licking, and lying patterns often appear before severe clinical symptoms. Individual identification is required to connect these health events to the correct animal.

Computer vision and deep learning provide a practical way to automate these observations using farm cameras. Convolutional Neural Networks (CNNs) can learn visual features from cattle images and videos, while transfer learning allows ImageNet-pretrained networks to be adapted to agricultural datasets. However, most existing systems are designed for only one task. A farm would therefore need separate models for BCS, behavior, lameness, and ID. This approach increases memory use, inference

latency, and deployment complexity.

This research addresses the above limitation by developing a unified multi-task deep learning framework. The model shares a visual backbone and then uses separate task heads for cattle monitoring outputs. The main idea is that cattle health tasks share visual cues: body shape helps BCS, posture helps behavior, limb movement helps lameness, and coat patterns or body structure help identification. By learning shared visual representations, the model can reduce computational cost and support edge-based farm deployment.

1.2 Rationale of the Study and Motivation

The motivation for this work comes from four major gaps observed in cattle monitoring research. First, many studies use private datasets, which makes reproduction and fair comparison difficult. Public reproducibility is important because a model that cannot be independently verified has limited scientific value. Therefore, this project prioritizes public datasets such as Dryad BCS, ScienceDB BCS, MmCows, CBVD-5, CattleLameness, Mendeley lameness data, and OpenCows2020.

Second, the majority of research focuses on isolated cattle monitoring problems. A separate model for each task wastes computation because all models repeatedly learn low-level visual features such as edges, contours, body texture, posture, and background separation. Multi-task learning can reduce this redundancy by sharing parameters across tasks.

Third, real farms require efficient deployment. A system running on farm cameras or edge devices cannot depend on several heavy models operating at the same time. Lightweight models such as MobileNetV3-Small and EfficientNetB0 are attractive because they provide a good balance between accuracy and computational efficiency.

Fourth, cattle monitoring involves both spatial and temporal information. A single image can be sufficient for BCS or ID, but lameness and behavior often depend on movement across time. This motivated the final spatiotemporal model, where a CNN extracts features frame by frame and an LSTM captures temporal sequence information.

1.3 Problem Statement

Despite progress in computer vision-based cattle monitoring, the field still lacks a reproducible and integrated framework that can handle BCS, behavior, lameness, and

individual identification together. Current systems usually require multiple independent models, each trained on different datasets and optimized for a single target. This creates four problems:

1. **Computational redundancy:** separate models repeatedly process the same cow image or video frame.
2. **Deployment complexity:** real farms would need to maintain multiple pipelines, checkpoints, and inference services.
3. **Limited cross-task learning:** related health signals are not used together, even though posture, gait, body shape, and identity are visually connected.
4. **Reproducibility limitations:** many existing studies rely on private datasets or highly controlled environments.

The central research problem is therefore: *How can a lightweight, reproducible, and unified deep learning framework be designed and evaluated for multiple cattle health and behavior monitoring tasks using publicly available datasets?*

1.4 Objectives

The main objective of this Pre-Thesis II work is to implement and evaluate a unified multi-task cattle monitoring framework. The specific objectives are:

1. To preprocess public datasets for BCS, behavior recognition, lameness detection, and cattle identification using consistent image size, normalization, augmentation, and cow-wise splitting.
2. To train single-task baseline models using five fixed backbone architectures: ResNet-18, MobileNetV3-Small, ResNet-50, DenseNet121, and EfficientNetB0.
3. To evaluate BCS using MAE and exact/within-one-class accuracy, behavior using Macro F1 and per-class accuracy, lameness using accuracy, F1, and AUC, and ID using Top-1 accuracy.
4. To select a shared backbone based on combined performance across tasks.
5. To implement a CBAM-enhanced multi-task model with separate heads for BCS, behavior, lameness, and ID.

6. To evaluate cross-dataset behavior generalization using an MmCows-trained model on CBVD-5.
7. To compare single-task and multi-task behavior to identify benefits and limitations of shared representation learning.

1.5 Methodology in Brief

The research pipeline begins with dataset collection and preprocessing. BCS images are collected from Dryad and ScienceDB. Behavior images are prepared from MmCows and cross-evaluated on CBVD-5. Lameness videos are cropped into cow-focused frames using YOLO-based preprocessing and then converted into 20-frame sequences. Cattle ID images are prepared from OpenCows2020.

Each dataset is resized to 224×224 pixels and normalized using ImageNet statistics. Training data uses random horizontal flipping and random rotation, while validation and test data do not use augmentation. For the behavior dataset, each class is capped at 3000 images to reduce severe class imbalance and speed up training.

The architecture uses an ImageNet-pretrained CNN backbone, followed by CBAM attention and task-specific heads. The BCS head uses CORAL ordinal regression, behavior uses focal loss, lameness uses binary cross-entropy, and ID uses softmax classification. Training begins with single-task baselines and then moves to the final multi-task model.

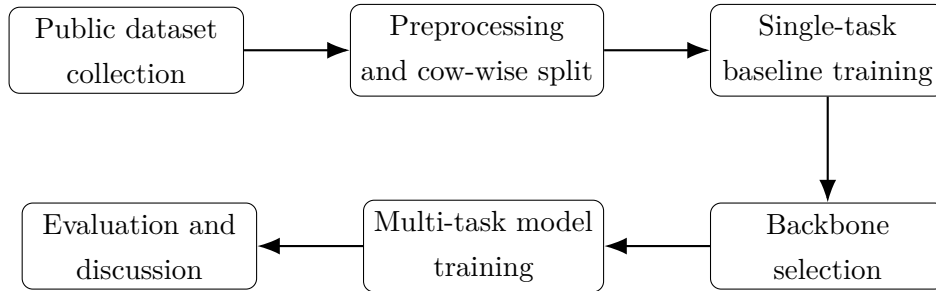


Figure 1.1: Methodology workflow for the Pre-Thesis II implementation.

1.6 Scope and Challenges

The scope of this research covers image- and video-based cattle monitoring for four tasks: BCS, behavior, lameness, and ID. The work emphasizes public datasets, reproducible preprocessing, single-task baselines, and a shared multi-task model. It does

not claim a final farm-ready commercial deployment; instead, it provides a working research prototype and performance evidence for the final thesis phase.

The main challenges are dataset heterogeneity, class imbalance, domain shift, limited lameness video samples, and destructive interference in multi-task training. Domain shift is especially visible in cross-dataset behavior evaluation, where a model trained on MmCows performs much worse on CBVD-5 because of differences in camera angle, background, annotation style, and farm setting. Multi-task training also introduces a trade-off: temporal tasks can improve when sequence information is shared, but static tasks may lose performance when the same backbone must support multiple competing objectives.

Chapter 2

Literature Review

2.1 Preliminaries

This research is based on several core concepts from computer vision, deep learning, and precision livestock farming. CNNs are used to extract visual features from cattle images. Transfer learning allows models pretrained on ImageNet to learn faster on relatively smaller agricultural datasets. Multi-task learning trains one shared representation for multiple related objectives. Attention mechanisms, such as CBAM, help the model focus on relevant channels and spatial regions. Ordinal regression is suitable for ordered labels such as BCS, where neighboring classes are more similar than distant classes. Temporal models such as LSTM networks are useful for lameness and behavior because movement patterns require information from multiple frames.

2.2 Review of Existing Research

Research on automated cattle monitoring has developed across several separate directions. For BCS, earlier work used RGB-D cameras, hand-engineered features, and later CNN-based models to estimate fat reserve or score category. Depth data is useful because BCS depends strongly on body shape and anatomical curvature, but public depth datasets are limited. Recent work has explored lightweight networks and attention mechanisms to improve efficiency and accuracy.

For lameness detection, existing methods use gait analysis, pose estimation, object detection, depth cameras, and temporal modeling. Lameness is difficult to classify from a single frame because the key signal is gait asymmetry over time. Therefore, sequence-based methods such as CNN-LSTM models are more suitable than pure image classifiers. However, public lameness datasets are much smaller than behavior or BCS datasets, which increases overfitting risk.

For behavior recognition, video-based systems classify behaviors such as walking, standing, feeding, drinking, licking, and lying. The MmCows dataset is valuable because it provides a large number of images, but it is severely imbalanced: common behaviors such as lying and standing appear much more often than rare behaviors such as licking and drinking. Loss functions such as focal loss and sampling strategies are therefore important.

For individual identification, vision-based cattle ID provides an alternative to physical tags and RFID systems. CNNs can use coat patterns, body markings, and head/body features to identify animals. However, ID can be affected by pose, occlusion, lighting, and dataset-specific farm conditions.

Finally, multi-task learning has been widely studied in deep learning, but its application to cattle monitoring remains limited. The literature suggests that related tasks can benefit from parameter sharing, but it also warns about negative transfer when tasks compete for the same shared representation. This project directly evaluates this trade-off in a cattle monitoring setting.

2.3 Summary of Key Findings

The literature review leads to five findings that motivate the proposed work:

1. Cattle monitoring tasks are usually studied separately, even though they share visual cues.
2. Public datasets are essential for reproducibility, but many existing studies use private farm data.
3. Lightweight backbones are needed for edge deployment in real farm environments.
4. Behavior and lameness require temporal or sequence-aware modeling for stronger generalization.
5. Cross-dataset evaluation is necessary because performance on one farm does not guarantee performance on another farm.

Table 2.1: Summary of literature directions and limitations.

Area	Common approach	Main limitation
BCS	RGB/RGB-D image classification or regression	Often single-task and sensitive to view/depth availability
Behavior	CNN or video-based classification	Severe class imbalance and domain shift
Lameness	Gait, pose, depth, or CNN-LSTM methods	Small datasets and need for temporal information
Cattle ID	Visual classification or metric learning	Sensitive to pose, lighting, and occlusion
MTL	Shared backbone with task-specific heads	Risk of negative transfer and task domination

Proposed Methodology and Implementation

3.1 Dataset Description

The Pre-Thesis II implementation uses multiple public datasets, each assigned to a specific task. The preprocessing strategy standardizes the image format so that all tasks can be connected to the same CNN backbone.

Table 3.1: Datasets used in the implementation.

Task	Dataset	Size	Description
BCS	Dryad BCS	5,923 images, 147 cows	DGE format with labels 2–6 mapped to ordinal indices 0–4
BCS	ScienceDB BCS	53,566 images, 10,898 cows	RGB images with labels 3.25–4.25 mapped to or- dinal indices 0–4
Behavior	MmCows	213,686 im- ages, 16 cows	RGB images with seven classes: walking, stand- ing, feeding head up, feeding head down, lick- ing, drinking, lying
Behavior	CBVD-5	2,000 balanced evaluation im- ages	Cross-dataset behavior evaluation set used for domain-shift testing
Lameness	Mendeley/CattleLameness	9,950 cropped frames across 50 videos	YOLO-cropped video frames converted to 20-frame sequences

Task	Dataset	Size	Description
ID	OpenCows2020	Preprocessed ID images	Individual cattle identification with EfficientNetB0 and CBAM

3.2 Preprocessing

All image inputs are resized to 224×224 pixels and normalized with ImageNet mean and standard deviation:

$$\mu = [0.485, 0.456, 0.406], \quad \sigma = [0.229, 0.224, 0.225]. \quad (3.1)$$

Training augmentation uses random horizontal flip with probability 0.5 and random rotation within $\pm 15^\circ$. Validation and test sets use no augmentation. A cow-wise split is used wherever cow identity information is available, ensuring that the same cow does not appear in both training and testing.

For MmCows behavior training, severe imbalance is reduced by capping each behavior class to a maximum of 3000 images. This creates a smaller and more balanced training set while retaining all classes. The raw behavior labels are mapped by subtracting one, so classes 1–7 become indices 0–6.

For BCS, label mapping is critical. Dryad integer labels are mapped as $\{2 : 0, 3 : 1, 4 : 2, 5 : 3, 6 : 4\}$. ScienceDB floating-point labels are mapped as $\{3.25 : 0, 3.5 : 1, 3.75 : 2, 4.0 : 3, 4.25 : 4\}$. This prevents label collapse and preserves ordinal structure.

3.3 Model Architecture

The proposed architecture follows a hard-parameter-sharing design. A shared backbone extracts visual features, CBAM attention refines those features, and separate task heads generate task-specific outputs. The backbone receives 224×224 three-channel input. For RGB datasets, the three channels are standard red, green, and blue. For Dryad, the three channels represent depth, grayscale, and edge information, but the format remains compatible with ImageNet-pretrained CNN backbones.

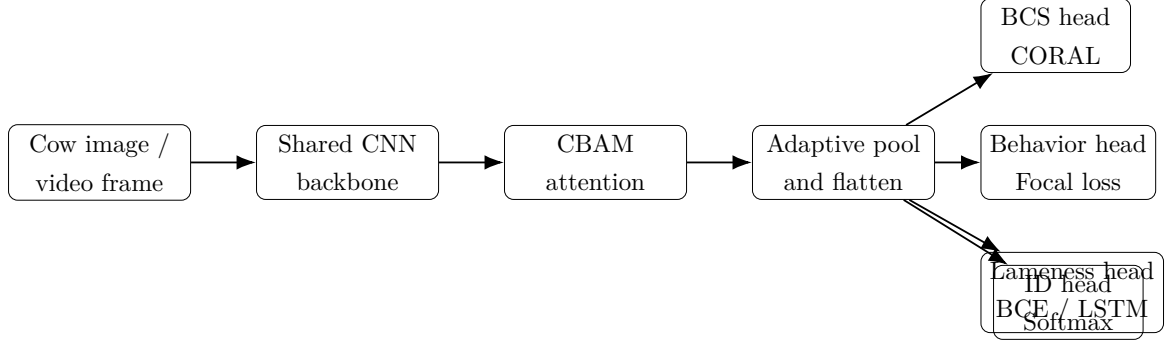


Figure 3.1: Conceptual overview of the shared multi-task architecture.

3.4 Backbone Candidates

Five model families were selected and assigned to the team members. The assignment is fixed across BCS and behavior tasks to enable fair comparison.

Table 3.2: Team-wise backbone assignment.

Member	Assigned backbone	Role in baseline comparison
Hasin Ishrak	ResNet-18	Lightweight residual baseline
Namira Abrar Haque	MobileNetV3-Small	Mobile edge baseline
Sanjida Akter Bithi	ResNet-50	Deeper residual baseline
Shouvik Banik	DenseNet121	Dense feature reuse baseline
Nusrat Lamya Faruk	EfficientNetB0	Efficient compound-scaled baseline

3.5 CBAM Attention Module

CBAM improves feature learning through two stages: channel attention and spatial attention. Channel attention decides what feature maps are important using average and maximum pooled descriptors. Spatial attention decides where the model should focus by combining channel-wise mean and maximum maps and applying a convolution.

Given a feature map F , CBAM can be summarized as:

$$F' = M_c(F) \otimes F, \quad (3.2)$$

$$F'' = M_s(F') \otimes F', \quad (3.3)$$

where M_c is channel attention, M_s is spatial attention, and $\otimes$ denotes element-wise

multiplication. In this project, CBAM is inserted between the backbone and the task heads.

3.6 Task-Specific Heads and Loss Functions

3.6.1 BCS Head with CORAL Loss

BCS is ordinal because the classes have a natural order. Predicting a neighboring score should be less severe than predicting a far-away score. Therefore, the BCS head uses CORAL ordinal regression with four output logits for five classes. A class prediction is obtained by counting how many sigmoid outputs exceed 0.5:

$$\hat{y} = \sum_{k=1}^{K-1} \mathbb{I}(\sigma(z_k) > 0.5). \quad (3.4)$$

3.6.2 Behavior Head with Focal Loss

Behavior recognition uses seven classes. The MmCows dataset is highly imbalanced, so focal loss is used to reduce the influence of easy majority-class examples. Focal loss is defined as:

$$L_{focal} = -\alpha(1 - p_t)^\gamma \log(p_t), \quad (3.5)$$

where $\gamma = 2$ and $\alpha = 0.25$ in the implementation.

3.6.3 Lameness Head

Lameness is treated as a binary classification problem: normal or lame. Spatial lameness uses CNN features from cropped frames. Spatiotemporal lameness uses 20-frame sequences and an LSTM to capture gait dynamics over time. This sequence-based approach is important because lameness depends on motion patterns rather than only body appearance.

3.6.4 ID Head

The cattle identification head uses a softmax classifier over cattle identity classes. The head is trained on OpenCows2020 using EfficientNetB0 with CBAM attention for 10 epochs.

3.7 Multi-Task Loss

The final multi-task model combines task-specific losses using a weighted sum:

$$L_{total} = w_1 L_{BCS} + w_2 L_{behavior} + w_3 L_{lameness} + w_4 L_{ID}. \quad (3.6)$$

The default weights are $w_1 = 0.35$, $w_2 = 0.35$, $w_3 = 0.15$, and $w_4 = 0.15$. BCS and behavior receive larger weights because they have larger public datasets and full baseline training. Lameness and ID are weighted lower because they are preliminary and have smaller dataset coverage.

3.8 Training Setup

Experiments were conducted on a Windows 11 research PC with Python 3.12.3, PyTorch 2.5.1+cu121, an NVIDIA RTX 4080 GPU with 16GB VRAM, and 64GB RAM. BCS experiments use Adam with learning rate 10^{-3} , StepLR scheduler, batch size 32, and 30 epochs. Behavior experiments use Adam with learning rate 10^{-3} , StepLR scheduler, batch size 128, AMP training, and early stopping with patience 10. Lameness and ID preliminary models were trained for 10 epochs using EfficientNetB0.

Chapter 4

Single-Task Implementation and Results

4.1 BCS Results

BCS was trained separately on Dryad and ScienceDB. The primary metric is MAE on ordinal indices 0–4. Lower MAE is better. Exact accuracy measures exact class match, while within-one-class accuracy measures whether the prediction is at most one ordinal class away from the true label.

Table 4.1: BCS baseline results on Dryad and ScienceDB.

Member	Backbone	Dryad MAE	Dryad Exact	Dryad ± 1	SciDB MAE	SciDB Exact	SciDB ± 1
Hasin	ResNet-18	0.8675	24.50%	88.75%	0.5800	54.81%	89.02%
Namira	MobileNetV3-Small	0.5250	61.50%	86.00%	0.7090	44.67%	86.47%
Bithi	ResNet-50	0.6300	47.75%	89.25%	0.6485	48.43%	88.20%
Shouvik	DenseNet121	0.5875	54.75%	86.50%	0.6292	49.51%	88.78%
Nusrat	EfficientNetB0	0.6175	52.50%	85.75%	0.5566	55.06%	90.50%

The Dryad results show that MobileNetV3-Small performs best on MAE and exact accuracy, despite being the smallest model. This suggests that the DGE representation can benefit from a lightweight model when training data is limited. ScienceDB results show that EfficientNetB0 performs best with MAE 0.5566 and within-one-class accuracy of 90.50%, suggesting stronger generalization on the larger RGB dataset.

4.2 Behavior Recognition Results

Behavior recognition was trained on MmCows with seven classes. The main metric is Macro F1-score because it treats every class equally and is more suitable for imbalanced datasets than overall accuracy.

Table 4.2: Behavior recognition baseline results on MmCows.

Member	Backbone	Epochs	Val Macro F1	Test Macro F1	Early stop
Hasin	ResNet-18	18	0.7241	0.7134	18
Namira	MobileNetV3-Small	30	0.7271	0.6810	N/A
Bithi	ResNet-50	22	0.7210	0.7037	22
Shouvik	DenseNet121	27	0.7371	0.7366	27
Nusrat	EfficientNetB0	24	0.7631	0.7445	24

EfficientNetB0 achieves the best validation and test Macro F1-score. DenseNet121 is the second-best behavior model. MobileNetV3-Small trains for all 30 epochs but has weaker test Macro F1, suggesting that smaller architecture alone does not guarantee generalization for behavior recognition.

Table 4.3: EfficientNetB0 behavior per-class accuracy on the MmCows test set.

Class	Test accuracy
Walking	59.78%
Standing	70.00%
Feeding head up	71.97%
Feeding head down	66.07%
Licking	83.96%
Drinking	83.01%
Lying	98.77%

The per-class results show that lying is the easiest class, while walking and feeding head down remain more difficult. This is expected because posture and background changes can make dynamic or transitional behavior classes more ambiguous.

4.3 Cross-Dataset Behavior Evaluation

The EfficientNetB0 behavior model trained on MmCows was evaluated on CBVD-5 with 2000 balanced images. The Macro F1-score was only 0.124517. Class-wise accuracy was 90.34% for standing, 8.39% for feeding head down, 2.99% for drinking, and 24.31% for lying.

This result shows strong domain shift. The model generalizes to standing postures but fails on several other classes because CBVD-5 differs in camera angle, farm back-

ground, annotation style, and visual distribution. This confirms that cross-dataset evaluation is essential before real deployment.

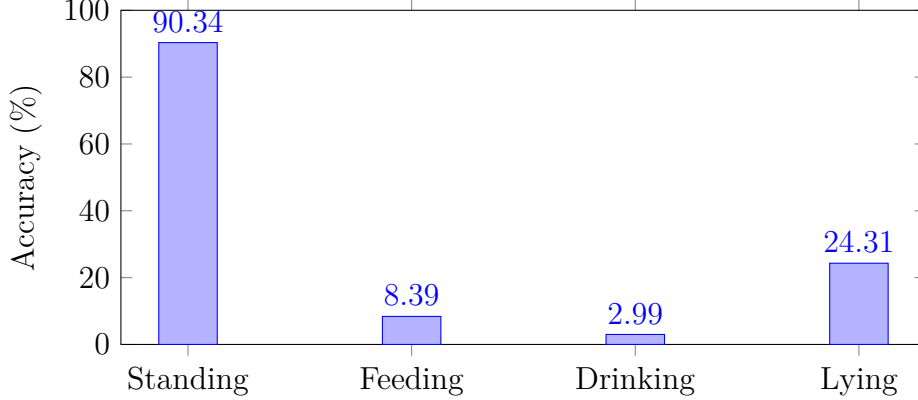


Figure 4.1: CBVD-5 cross-dataset behavior accuracy by class.

4.4 Lameness Results

Lameness was implemented in both spatial and spatiotemporal forms. The spatial EfficientNetB0 model reached a test AUC of 0.982942 and test accuracy of 93.05%. The EfficientNetB0-LSTM spatiotemporal model used 20-frame sequences and reached test AUC 0.8400 and test accuracy 80.00% under the default 0.5 threshold.

Table 4.4: Lameness detection results.

Model	Epochs	Test AUC	Test accuracy	Test F1
EfficientNetB0 spatial	10	0.9829	93.05%	0.9483
EfficientNetB0-LSTM sequence	15	0.8400	80.00%	0.8000
ResNet18-LSTM sequence	15	0.9600	60.00%	0.7143

The results show that the spatial model performed strongly on the cropped-frame setting. The spatiotemporal model remains important for final multi-task deployment because it can represent motion and is more appropriate for gait-based reasoning. The difference between AUC and accuracy also indicates that threshold calibration may be required for sequence-level lameness classification.

4.5 Cattle Identification Results

Cattle ID was trained on OpenCows2020 using EfficientNetB0 with CBAM attention for 10 epochs. The model achieved 87.06% validation Top-1 accuracy and 86.49% test

Top-1 accuracy.

Table 4.5: OpenCows2020 ID result.

Backbone	Epochs	Final train loss	Val Top-1	Test Top-1
EfficientNetB0 + CBAM	10	0.550795	87.06%	86.49%

4.6 Backbone Selection

Backbone selection considers BCS MAE, behavior Macro F1, lameness AUC, and ID accuracy. EfficientNetB0 has the strongest average profile because it obtains the best behavior Macro F1, the best ScienceDB BCS MAE, strong lameness performance, and completed ID results. Therefore, EfficientNetB0 is selected as the shared backbone for the final multi-task model.

Table 4.6: Backbone comparison summary.

Backbone	Dryad BCS MAE	SciDB BCS MAE	Behavior F1	Lameness AUC	ID Top-1
ResNet-18	0.8675	0.5800	0.7134	0.8400 (ST)	Pending
MobileNetV3-Small	0.5250	0.7090	0.6810	Pending	Pending
ResNet-50	0.6300	0.6485	0.7037	Pending	Pending
DenseNet121	0.5875	0.6292	0.7366	Pending	Pending
EfficientNetB0	0.6175	0.5566	0.7445	0.9829	86.49%

Multi-Task Model and Discussion

5.1 Final Multi-Task Model

The final implementation uses EfficientNetB0 as the shared backbone with CBAM attention. The spatial multi-task version evaluates the four heads directly, while the spatiotemporal version uses 20-frame sequences and temporal modeling for sequence-sensitive outputs. The trained checkpoint is stored as `multitask_temporal_best.pth`.

The final spatiotemporal multi-task results are summarized in Table 5.1.

Table 5.1: Single multi-task model evaluation results.

Task	Spatial multi-task	Spatiotemporal multi-task
BCS	MAE 0.6919, Exact 0.4285	MAE 0.7849, Exact 0.3907
Behavior	Macro F1 0.3739	Macro F1 0.4775
Lameness	Accuracy 0.9432, AUC 0.9884	Accuracy 1.0000, AUC 1.0000
Cow ID	Accuracy 0.9597	Accuracy 0.9718

The spatiotemporal model improves behavior over the spatial multi-task version and achieves perfect lameness accuracy and AUC on the evaluated split. It also improves ID accuracy to 97.18%. However, BCS performance becomes weaker compared to the best single-task BCS baselines. This shows a common multi-task trade-off: sharing one backbone reduces model cost and improves some tasks, but can cause negative transfer for tasks that require different feature emphasis.

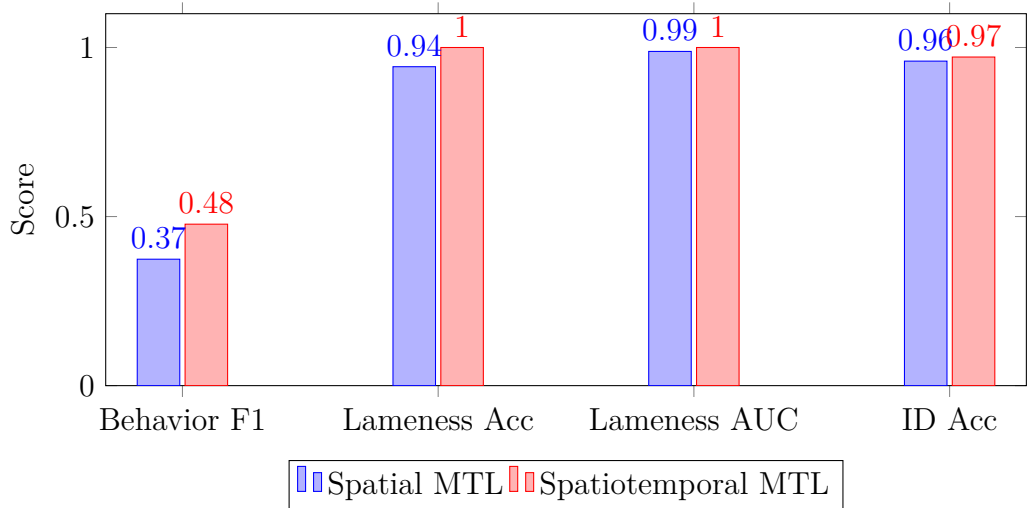


Figure 5.1: Comparison of spatial and spatiotemporal multi-task results.

5.2 Single-Task versus Multi-Task Comparison

The single-task baselines remain stronger for BCS and behavior. For example, the best ScienceDB BCS single-task MAE is 0.5566, while the final spatiotemporal multi-task BCS MAE is 0.7849. The best single-task behavior Macro F1 is 0.7445, while the spatiotemporal multi-task behavior F1 is 0.4775. This reduction indicates that the multi-task model needs further loss balancing, task-specific adapters, or staged fine-tuning.

At the same time, the multi-task model provides major deployment advantages. Instead of running four full models, a single backbone extracts features once. This reduces memory cost and inference overhead. It also produces a unified output set for farm monitoring: BCS, behavior, lameness, and cow identity. For real-time use, this integrated inference is more practical than separate models.

5.3 Ablation Studies

Seven ablation directions were identified. Some are completed and some remain for the final thesis stage.

Table 5.2: Ablation study status.

Ablation	Status	Purpose / finding
CORAL vs Cross-Entropy for BCS	Planned	Measure whether ordinal loss improves BCS MAE
CBAM vs no CBAM	Planned	Measure contribution of attention module
RGB only vs DGE / depth-aided input	Planned	Evaluate benefit of structural depth cues
Focal loss vs Cross-Entropy for behavior	Planned	Measure behavior imbalance handling
MmCows to CBVD-5 cross-dataset evaluation	Complete	Macro F1 0.124517; strong domain shift observed
Backbone comparison	Complete	EfficientNetB0 selected as best average backbone
Single-task vs multi-task	Complete	MTL improves deployment and temporal tasks but reduces BCS/behavior scores

5.4 Real-Time Inference Concept

The final deployment concept uses a rolling 20-frame sequence from a farm video stream. For BCS and ID, predictions can be averaged or voted across frames to reduce noise. For behavior and lameness, temporal features are passed through the LSTM to capture movement patterns. A real-time rolling sequence visualizer was successfully deployed and tested on a demo video, confirming that the multi-output inference pipeline can run as an integrated prototype.

5.5 Limitations

The current implementation has several limitations. First, the multi-task model does not yet outperform all single-task baselines. Second, lameness training uses a small video dataset, which can inflate validation performance and make threshold calibration important. Third, cross-dataset behavior evaluation shows severe domain shift, meaning the model still needs stronger domain adaptation or multi-farm training. Fourth, some ablation studies such as CORAL vs Cross-Entropy and CBAM vs no CBAM are planned but not yet completed. Fifth, the ID and lameness tasks are

preliminary compared to BCS and behavior.

Conclusion and Future Work

6.1 Conclusion

This Pre-Thesis II report presents the implementation progress of a unified multi-task deep learning framework for cattle health and behavior monitoring. The project moved beyond the conceptual Pre-Thesis I stage by preprocessing datasets, training five baseline backbones, evaluating BCS and behavior models across the full team, implementing lameness and ID models, completing cross-dataset behavior evaluation, and training a final spatiotemporal multi-task model.

The results show that EfficientNetB0 is the strongest overall backbone. It achieves the best ScienceDB BCS MAE, the best behavior Macro F1-score, strong lameness AUC, and completed ID performance. The final spatiotemporal multi-task model shows the feasibility of unified cattle monitoring, achieving 100.00% lameness accuracy and 97.18% ID accuracy. However, BCS and behavior scores decrease compared to single-task baselines, confirming the need for additional multi-task optimization.

Overall, the work demonstrates that a single shared framework can support multiple cattle monitoring tasks using public datasets and lightweight architectures. It also reveals important research challenges, especially domain shift, class imbalance, task interference, and temporal modeling.

6.2 Future Work

The final thesis phase will focus on improving multi-task performance and robustness. Planned future work includes:

1. Completing CORAL vs Cross-Entropy, CBAM vs no CBAM, RGB vs DGE, and focal loss vs Cross-Entropy ablation studies.

2. Fine-tuning multi-task loss weights and testing equal weighting as a baseline.
3. Adding task-specific adapters or lightweight task gates to reduce negative transfer.
4. Improving behavior generalization using domain adaptation and multi-farm augmentation.
5. Calibrating lameness thresholds and evaluating more video sequences.
6. Expanding ID evaluation to additional cattle identity datasets such as Cows2021.
7. Testing model latency, parameter count, and memory use for edge deployment.
8. Exploring temporal modeling for behavior recognition, not only lameness.
9. Considering future modalities such as thermal imaging when public data becomes available.
10. Investigating federated learning for privacy-preserving training across multiple farms.

Appendix A

Implementation Details

A.1 Project Directory Summary

The verified project structure includes dataset folders for BCS, behavior, lameness, and ID; workspace folders for each team member; and a final model folder containing the completed multi-task checkpoint. The main final checkpoint is `D:\T25301094 P2\final_models\multitask_temporal_best.pth`.

A.2 Important Reproducibility Rules

The implementation follows several rules to avoid common errors: label mapping is applied inside dataset loading; ScienceDB float labels are not cast to integers; model feature dimension is detected dynamically using a dummy forward pass; `torch.compile` is avoided because it may fail on Windows with adaptive pooling; and checkpoints are loaded using `weights_only=True` where applicable.

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